

Experiment 13: Alpha-Beta Pruning Algorithm

Aim

To implement the Alpha-Beta Pruning algorithm using Python.

Objective

To improve the efficiency of the Minimax algorithm by eliminating unnecessary branches.

Algorithm

1. Initialize Alpha and Beta values.
 2. Evaluate game states.
 3. Update Alpha and Beta values.
 4. Prune branches when possible.
 5. Display the final result.
-

Flowchart

Start

|

Generate Game Tree

|

Evaluate Nodes

|

Prune Branch?

/ \

Yes No

|

|

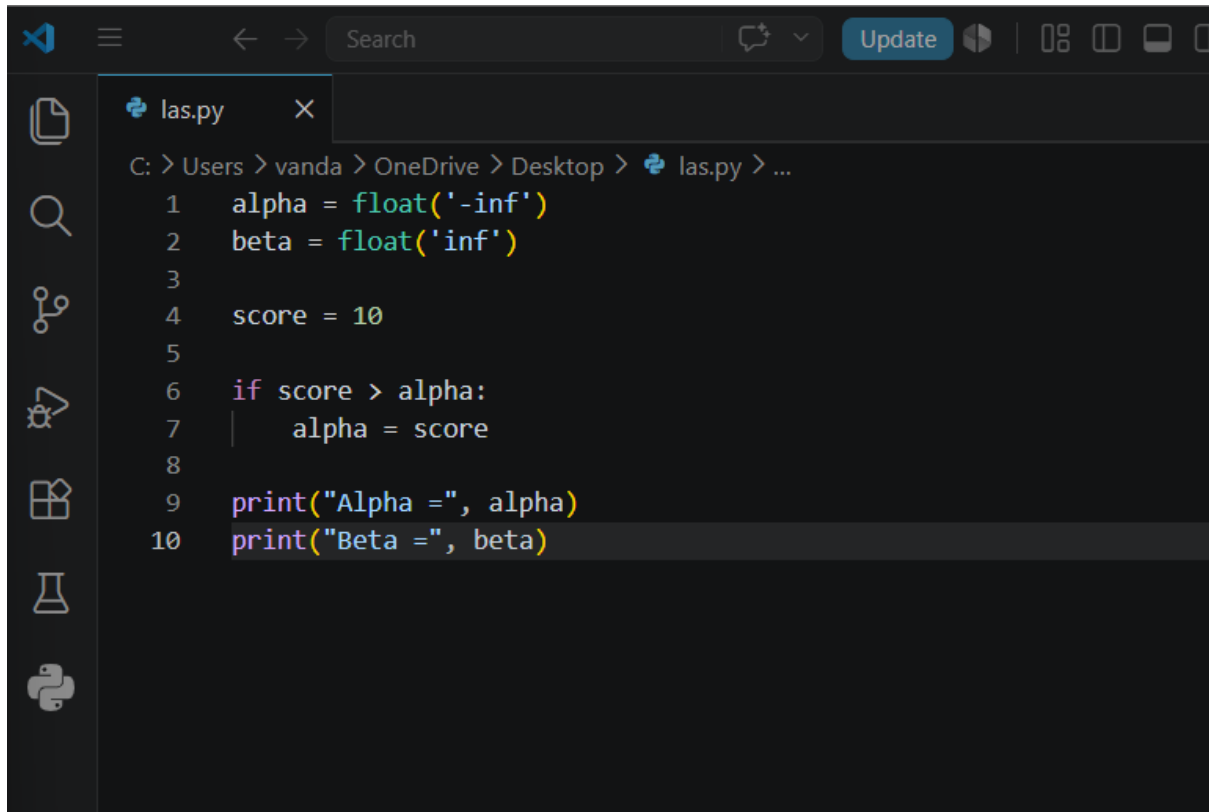
Skip Continue

|

Display Result

|
Stop

Code



```
1  alpha = float('-inf')
2  beta = float('inf')
3
4  score = 10
5
6  if score > alpha:
7      alpha = score
8
9  print("Alpha =", alpha)
10 print("Beta =", beta)
```

Output:

```
PS C:\Users\vanda\onedrive\desktop> PYTHON las.py
Alpha = 10
Beta = inf
PS C:\Users\vanda\onedrive\desktop> |
```

Result

The Alpha-Beta Pruning algorithm was executed successfully.

Conclusion

Alpha-Beta Pruning reduces the number of nodes evaluated, making the Minimax algorithm faster.